POWER ANALYSIS REPORT

Statistical Power Analysis

Multiple Linear Regression

STATISTICAL POWER

99.70%

Power = 0.9970 (Type II error rate = 0.0030)

INPUTS

Test	**Multiple Linear Regression**
Mode	**power**
Significance level	**0.05**
Effect size (f^2)	**0.15**
Total observations	**150**
Predictors	**1**

DESIGN JUSTIFICATION

A multiple regression power analysis used alpha = 0.05, f^2 = 0.15, 1 predictor coefficients, excluding the intercept. With total observations = 150, estimated power was 99.7%. This is a design calculation for the specified effect, not evidence for or against a hypothesis.

HOW TO CITE

StatsCalculators Team. (2026). Sample Size & Power Analysis Calculator. StatsCalculators. Retrieved November 9, 2026 from https://www.statscalculators.com/calculators/hypothesis-testing/sample-size-and-power-analysis-calculator

EFFECT INTERPRETATION

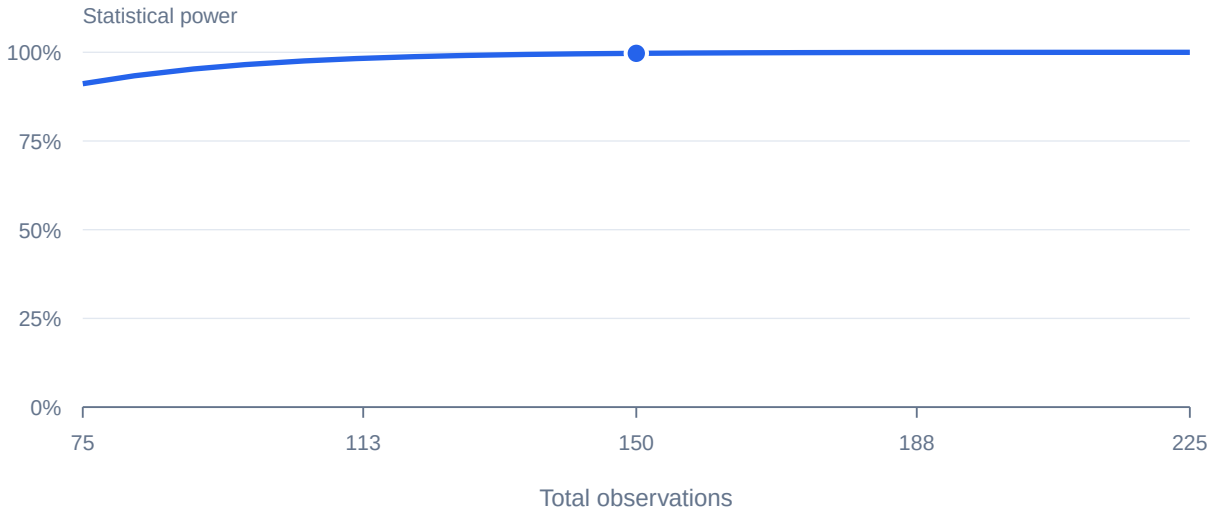
Effect to detect: 0.15 (Cohen's f^2). Judge practical importance in the context of your study.

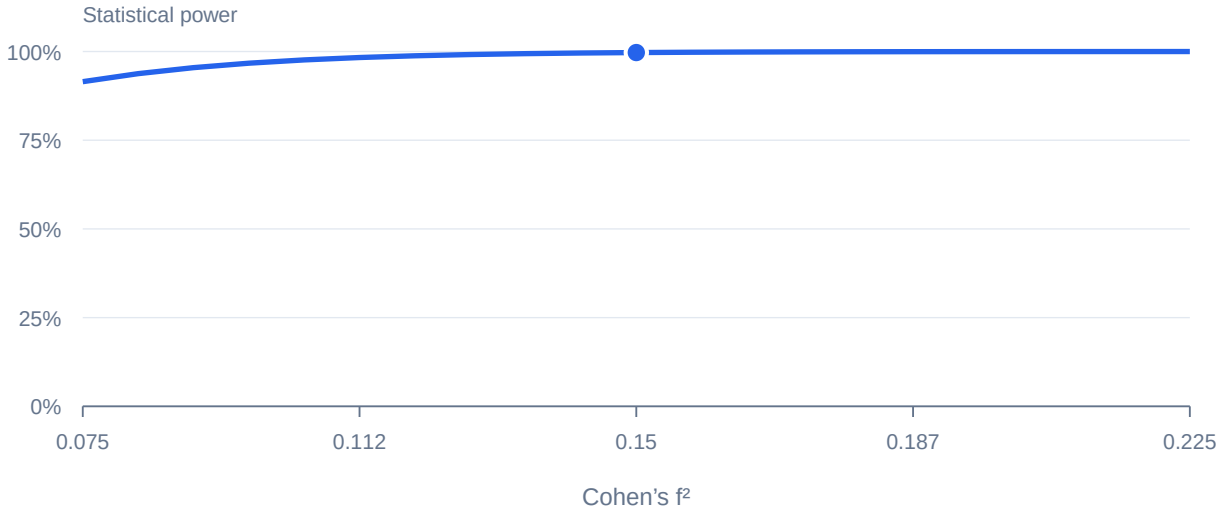
Power analysis charts

All other assumptions stay fixed. The blue dot marks your selected design.

How sample size changes power

Selected design: 150; achieved power: 99.70%.



How effect size changes power

Selected design: 0.15; achieved power: 99.70%.